KOSZUL DUALITY AND CATEGORICAL ARCHIMEDEAN LOCAL LANGLANDS

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1. Kazhdan–Lusztig inversion formula

Somehow the categorification of local Langlands correspodence started off as interpreting the Kazhdan–Lusztig inversion formula. We recall the setup. Namely, consider the block $\mathcal{O}_0$ of the category of representations of a complex semisimple Lie algebra $\mathfrak{g}$ which contains the trivial representation. The simple objects are the simple highest weight modules M_w of highest weight $-w(\rho)-\rho$, for the half sum of positive roots ρ and $w \in W$. This is the irreducible quotient of the Verma module M_w of highest weight $-w(\rho)-\rho$ which is universal among highest weight modules. The **Kazhdan–Lusztig polynomial** $P_{y,w}(q) \in \mathbb{Z}[q]$ for $y, w \in W$ encodes the multiplicity of Verma module, in that

$$[M_w] = \sum_{y \leq w} P_{w_0 w, w_0 y}(1)[L_y].$$

The Kazhdan–Lusztig inversion formula says that the inversion is also expressible in terms of K-L polynomials, namely

$$[L_w] = \sum_{y \leq w} (-1)^{\ell(w)-\ell(y)} P_{y,w}(1)[M_y].$$

[BGS] found that the categorification of this is that

$$D^b(\mathcal{O}_0) \cong D^b_{(B^\vee)}(G^\vee/B^\vee) \qquad \text{(Categorification of local Langlands over } \mathbb{C}\text{)},$$

where the right hand side is the bounded derived category of $B^\vee$-equivariant constructible sheaves on $G^\vee/B^\vee$, and this is an instance of **Koszul duality**. We will see what Koszul duality means shortly. Soergel then generalized this picture to real group representations, which we will also see. To at least superficially understand what this $\cong$ is, we translate the left-hand side to a geometric picture using Beilinson–Bernstein localization;

$$D^b_{(B)}(G/B) \cong D^b(\mathcal{O}_0) \cong D^b_{(B^\vee)}(G^\vee/B^\vee).$$

Then L_w corresponds to the **IC sheaf**, which is a simple perverse sheaf corresponding to the Schubert cell associated with w, and M_w corresponds to the **standard object**. In terms of $i_w : BwB/B \hookrightarrow G/B$,

$$L_w = i_{w!*}k[\ell(w)], \quad M_w = i_{w!}k[\ell(w)].$$

Similar definition $N_w = i_{w*}k[\ell(w)]$ corresponds to the **dual Verma module** $N_w = M_w^\vee$.

Now the "Koszul duality" $D^b_{(B)}(G/B) \cong D^b_{(B^\vee)}(G^\vee/B^\vee)$ is such that $L_w \mapsto N_{w^{-1}w_0}$, and $M_w \mapsto L_{w^{-1}w_0}$. Somehow you can now imagine that the roles of L and M are "flipped."

2. Koszul duality basics

So why is this called Koszul duality? Koszul duality is basically a generalization of

$$D^b(\mathrm{Mod}_{S(V)}) \cong D^b(\mathrm{Mod}_{\wedge^*(V^\vee)}).$$

This is usually called the BGG correspondence. Let us briefly recall what this is. Let $M = \oplus M_d$ be a graded $S(V)$-module. Then, define $R(M)$ to be the complex

$$\cdots \xrightarrow{\phi} \mathrm{Hom}_K(\wedge^*(V^\vee), M_d) \xrightarrow{\phi} \mathrm{Hom}_K(\wedge^*(V^\vee), M_{d+1}) \xrightarrow{\phi} \cdots,$$

$$\phi(\alpha)(e) = \sum_i v_i \alpha(\widetilde{v}_i \wedge e),$$

where v_i and $\widetilde{v}_i$ are dual bases of V and $V^\vee$. This being a complex is basically equivalent to that the induced morphisms $V \otimes M_d \to M_{d+1}$ satisfy commutativity and associativity so that it gives an $S(V)$-module structure on M. This gives an equivalence of categories between graded left $S(V)$-modules and certain category of **complexes of $\wedge^* V^\vee$-modules**. This can be upgraded to an equivalence between derived categories by taking the total complex of R.

So, Koszul duality is really a "derived" phenomenon that involves two different t-structures encoded in terms of two different "gradings." Let us develop a more general theory of Koszul duality.

Definition 2.1 ([BGS, Def. 1.1.2])**.** A Koszul ring is a positively graded ring $A = \oplus_{j \geq 0} A_j$ such that

(1) A_0 is semisimple (the category of A-modules is semisimple, i.e. every short exact sequence splits),
(2) A_0 admits a graded projective resolution of graded left A-modules

$$\cdots \to P^2 \to P^1 \to P^0 \twoheadrightarrow A_0,$$

such that P^i is generated by its degree i component, $P^i = AP^i_i$.

Example 2.2. The name of Koszul comes from the fact that the Koszul complex of a regular ring shows that a regular ring is Koszul. For example, for $A = k[x, y]$ with usual grading,

$$0 \to xyA \xrightarrow{f \mapsto (f, -f)} (xA) \oplus (yA) \xrightarrow{(a,b) \mapsto a+b} A \twoheadrightarrow k \to 0,$$

shows that A is Koszul. More generally, a polynomial ring, or in other words a symmetric algebra is Koszul. Similarly, an exterior algebra is also Koszul.

Koszulity is something really close to semisimplicity.

Proposition 2.3. *Let A be a Koszul k-algebra for semisimple k.*

(1) For any two pure modules M, N of weights m, n (namely, concentrated in one degree of grading), $\mathrm{Ext}^i_A(M, N) = 0$ unless $i = m - n$.
(2) A is quadratic, namely $A = T(V)/R$ for a k-vector space V and $R \subset V \otimes V$.

Definition 2.4. Let $A = T(V)/R$ be a Koszul k-algebra. Then its **Koszul dual** $A^!$ is $T(V^*)/R^\perp$, where $R^\perp \subset V^* \otimes V^* = (V \otimes V)^*$ consists of those on which the elements of R vanish.

Now here is Koszul duality.

Proposition 2.5. *Let A be Koszul.*

(1) $A^!$ is Koszul.
(2) $A^{!!} = A$.
(3) $\mathrm{Ext}^*_A(k, k) = (A^!)^{\mathrm{op}}$.
(4) There is a natural equivalence of categories between $D^+(A - \mathrm{mod})$ and $D^-(A^! - \mathrm{mod})$.

Proof. The way that $A^! = T(V^*)/R^\perp$ is realized gives a graded projective resolution of k as an A-module,

$$\cdots \to \mathrm{Hom}_k(A^!_2, A) \to \mathrm{Hom}_k(A^!_1, A) \to A \to k \to 0,$$

$$(df)(a) = \sum_i f(a\widehat{v}_i)v_i,$$

where $\{v_i\}$ is a basis of V and $\{\widehat{v}_i\}$ is the dual basis of V^*. This complex ("Koszul complex") can be thought as $\mathrm{Hom}_A(A \otimes A^!, A)$ with the differential $a \otimes b \mapsto \sum_i av_i \otimes \widehat{v}_i b$. In general, the equivalence of categories should be constructed as

$$M \in D^+(\mathrm{Mod}_A) \mapsto \mathrm{Hom}_A(A \otimes A^!, A) \in D^-(\mathrm{Mod}_{A^!}),$$

$$N \in D^-(\mathrm{Mod}_{A^!}) \mapsto A \otimes A^! \otimes_{A^!} N \in D^+(\mathrm{Mod}_A).$$

□

Remark 2.6. The duality does not given an equivalence of unbounded derived category. For example, let $A = k[x]$ and $A^! = k[\epsilon]/\epsilon^2$. Then, $A \in \mathrm{Mod}_A$ is sent to $0 \to k[x] \xrightarrow{x} k[x] \to 0 \cong k\langle 1\rangle[-1]$. Note that A is compact in $D(\mathrm{Mod}_A)$, but k is not compact in $D(\mathrm{Mod}_{A^!})$. Indeed, k is resolved in $D(\mathrm{Mod}_{A^!})$ (here we use unboundedness in the wrong direction) as

$$\cdots \to k[\epsilon]\langle 2\rangle \xrightarrow{\epsilon} k[\epsilon]\langle 1\rangle \xrightarrow{\epsilon} k[\epsilon],$$

so $\mathrm{Hom}_{D(\mathrm{Mod}_{k[\epsilon]})}(k, k[\epsilon]\langle n\rangle[n]) = 0$, whereas $\mathrm{Hom}_{D(\mathrm{Mod}_{k[\epsilon]})}(k, \oplus_{n\geq 1} k[\epsilon]\langle n\rangle[n])$ is k.

Now why Kazhdan–Lusztig inversion formula gives rise to Koszul duality is via the following.

Proposition 2.7 (Numerical Koszulity criterion)**.** *If k is a semisimple F-algebra for a field F such that $k = \bigoplus_{x\in W} F1_x$ for pairwise orthogonal idempotents indexed by a finite set W, then for a positively graded ring A which is an F-algebra with $A_0 = k$, the Hilbert polynomial of A is a $|W| \times |W|$-matrix*

$$P(A, t)_{x,y} = \sum_{i\geq 0} t^i \dim(1_x A_i 1_y).$$

Then, for a noetherian A, A is Koszul iff $P(A, t)P(A^!, -t)^T = 1$.

Proof. We just prove one side. If A is Koszul, because of Koszul complex, for any $n > 0$,

$$\sum_{i=0}^{n} (-1)^i \dim(1_x \operatorname{Hom}_k(A^!_{n-i}, A_i) 1_y) = 0$$

(Euler characteristic). □

Indeed, Soergel proved that $\mathcal{O}_0$ is equivalent to the category of nilpotent $\operatorname{Ext}_{B^\vee}(G^\vee/B^\vee)$-modules, where $\operatorname{Ext}_{B^\vee}(G^\vee/B^\vee)$ is the self-Ext-algebra of $\mathcal{L} \in \mathcal{P}_{B^\vee}(G^\vee/B^\vee)$, the category of $B^\vee$-equivariant perverse sheaves, defined as the sum of all simple objects. This shows not only that the derived categories are equivalent but also the corresponding abelian categories are in some sense Koszul-dual to each other.

We will see that a similar pattern holds for local Langlands for real reductive groups.

3. Archimedean local Langlands

We review what archimedean local Langlands is. This is a consequence of two theories, Langlands classification and Knapp–Zuckerman theory.

Theorem 3.1 (Langlands classification)**.** *Every irreducible $(\mathfrak{g}, K)$-module is the unique irreducible quotient of the parabolic induction of a tempered representation. This gives a parametrization of irreducible $(\mathfrak{g}, K)$-modules.*

Theorem 3.2 (Knapp–Zuckerman)**.** *Every tempered irreducible $(\mathfrak{g}, K)$-module is **basic**, meaning that it is a constituent of the parabolic induction of a (limit of) discrete series. This gives a parametrization; you know exactly how parabolic induction reduces, etc.*

Now limits of discrete series are classified by the weight and a Weyl chamber (with fixed positive system of roots for $\mathfrak{k}$). In contrast to p-adic case, real case is completely classified because

- one has simple classification of discrete series = supercuspidals,
- and one knows how principal series reduces.

4. ABV geometric parameter space and Soergel's conjecture

A formula similar to Kazhdan–Lusztig inversion formula holds in the context of representation of real groups under the name of **Vogan duality**, or Vogan's character-multiplicity formula. It is still the case by Langlands classification that an irreducible object is a unique irreducible quotient of a **standard object**, which is analogous to Verma module. Vogan duality is a duality between the multiplicity of irreducible representation in a standard module (principal series) and the number that appears in the expression of the character of irreducible as a linear combination of standards, just like in the case of Kazhdan–Lusztig inversion formula. However, there are a few subtleties compared to the case of Kazhdan–Lusztig theory for category $\mathcal{O}$, and they are really manifestation of what happens for general local Langlands.

- In the category $\mathcal{O}$ setting, the irreducible representations are conveniently characterized by highest weights. This is no longer true, and one really needs to consider representations of two different groups.

  Example 4.1. Let $G = \mathrm{SL}_2(\mathbb{R})$ and consider the block containing the trivial representation π_{triv}. This has two other irreducibles, π_+, π_-, two discrete series. One knows

$I_{\mathrm{triv}} \twoheadrightarrow \pi_{\mathrm{triv}}$ whereas $I_\pm = \pi_\pm$. Also, $I_{\mathrm{triv}} = \pi_{\mathrm{triv}} + \pi_+ + \pi_-$. So $\pi_{\mathrm{triv}} = I_{\mathrm{triv}} - I_+ - I_-$, and the character matrix is

$$\begin{pmatrix} 1 & -1 & -1 \\ & 1 & 0 \\ & & 1 \end{pmatrix}.$$

Thus, presumably the dual side should have multiplicity matrix

$$\begin{pmatrix} 1 & & \\ 1 & 1 & \\ 1 & 0 & 1 \end{pmatrix},$$

because sign is changed to be positive (multiplicity is positive). However there is only one principal series splitting into a sum of two irreducibles, so this is not realizable over $\mathrm{SL}_2(\mathbb{R})$.

Instead, one should look at its dual, $\mathrm{PGL}_2(\mathbb{R})$ (or $\mathrm{SL}_2^{\pm}(\mathbb{R})$ if you want). There the block of the trivial representation has one discrete series, π', which combines both π_+ and π_-. On the other hand, now we have **two finite dimensional representations**, namely the trivial representation and the sign representation. One readily checks that $I'_{\mathrm{triv}} = \pi'_{\mathrm{triv}} + \pi'$ and $I'_{\mathrm{sgn}} = \pi'_{\mathrm{sgn}} + \pi'$.

- Over the complex numbers, there is no Vogan L-packet issue. Here we need Vogan L-packet.

Theorem 4.2 (Vogan's character-multiplicity formula, [Vo1, Thm. 1.15])**.** *Let $G^{\mathbb{C}}$ be a complex connected reductive group, and G a real form of $G^{\mathbb{C}}$. Fix a block $\{\pi_1, \cdots, \pi_r\}$ of irreducible representations of G, having the same infinitesimal character as some finite dimensional representation of G. Write $\widehat{G}^{\mathbb{C}}$ for the complex simply connected semisimple Langlands dual group. Then there is a real form $\widehat{G}$ of $\widehat{G}^{\mathbb{C}}$ and a block $\{\rho_1, \cdots, \rho_r\}$ of irreducible representations of $\widehat{G}$ such that*

$$\pi_i = \sum_j \pm[\widetilde{\rho}_j : \rho_i]\widetilde{\pi}_j,$$

$$\rho_i = \sum_j \pm[\widetilde{\pi}_j : \pi_i]\widetilde{\rho}_j,$$

where π_i is the Langlands quotient of $\widetilde{\pi}_i$, etc. In this duality, discrete series is sent to the Langlands quotient where the Levi is a split group.

Note that even though this is a duality, the machine does not spit out π_i's and G when you plug ρ_i's and $\widehat{G}$.

Motivated by the above picture for KL, Adams–Barbasch–Vogan tried to relate the category of block of G and the category of equivariant perverse sheaves/D-modules. The idea is that the real case is the $\Gamma = (\mathbb{Z}/2\mathbb{Z})$-invariant version of the complex version. In the case when G is a complex Lie group, the geometric space was $\widehat{B}\backslash\widehat{G}/\widehat{B}$. Given a real form G of a complex group $G_{\mathbb{C}}$, we have an antiholomorphic involution γ on $G_{\mathbb{C}}$. This induces an involution η on $\widehat{G}$, defined as follows. Let us fix the pinning of $G_{\mathbb{C}}$. On the character lattice X of $G_{\mathbb{C}}$, we define $\eta : X \to X$ to be

$$(\eta(\lambda))(t) = \overline{\lambda(\gamma(t))},$$

for all $t \in T$. Then η on $\widehat{G}$ is defined to be induced from the adjoint $\eta : X^\vee \to X^\vee$ to $\eta : X \to X$. We would like to conjecture something like

$$D^b(\mathrm{Rep}_G) \overset{?}{\cong} D^b((\widehat{B}\backslash\widehat{G}/\widehat{B})^{\eta=1}).$$

There are two problems:

(1) The left hand side should consider several real forms of $G_\mathbb{C}$ at once, as we know.
(2) The involution on the right hand side should flip the left and right $\widehat{B}$-quotient. Now the quotient $\widehat{B}\backslash\widehat{G}/\widehat{B}$ is really identified with

$$\widehat{B}\backslash\widehat{G}/\widehat{B} = \widehat{G}\backslash(\widehat{G}/\widehat{B} \times \widehat{G}/\widehat{B}) = B\widehat{B} \times_{B\widehat{G}} B\widehat{B}.$$

The correct involution η is actually the η induced from $\eta : \widehat{G} \to \widehat{G}$ composed with the involution on $B\widehat{B} \times_{B\widehat{G}} B\widehat{B}$ "flipping the two coordinates."

I claim that at least this phenomenon is visible from the corrected version right hand side. First we work with derived stacks, which means we treat everything in derived sense to the extreme. This makes sense because already we used equivariant derived category of constructible sheaves and that has good categorical compatibility only on the level of derived stacks. Otherwise it's an extremely subtle object. We first need to know what derived fixed point means. It means

$$Y^\Gamma = \mathrm{Map}_{B\Gamma}(B\Gamma, Y/\Gamma).$$

Indeed, a Γ-fixed point at least gives rise to a $B\Gamma$-morphism $B\Gamma \to Y/\Gamma$. Now what is $(\widehat{B}\backslash\widehat{G}/\widehat{B})^\Gamma$? In terms of $\widehat{G}\backslash(\widehat{G}/\widehat{B} \times \widehat{G}/\widehat{B})$, in classical terms the Γ-fixed points should be $\widehat{G}^\Gamma\backslash\widehat{G}/\widehat{B}$ as one can identify the two factors of $\widehat{G}/\widehat{B}$. Thus it is reasonable to believe that the derived version would yield

$$(\widehat{B}\backslash\widehat{G}/\widehat{B})^\Gamma = (B\widehat{G})^\Gamma \times_{B\widehat{G}} B\widehat{B}.$$

So what is $(B\widehat{G})^\Gamma$? From the definition,

$$(B\widehat{G})^\Gamma = \mathrm{Map}_{B\Gamma}(B\Gamma, (B\widehat{G})/\Gamma) = \mathrm{Map}_{B\Gamma}(B\Gamma, B(\widehat{G} \rtimes \Gamma)),$$

where $\widehat{G} \rtimes \Gamma$ is the natural semidirect product. Thus

$$(B\widehat{G})^\Gamma = \{\Gamma \to \widehat{G} \rtimes \Gamma \mid \Gamma \to \widehat{G} \rtimes \Gamma \twoheadrightarrow \Gamma \text{ is the identity}\}/\widehat{G},$$

where the action is via conjugation. Thus

$$(B\widehat{G})^\Gamma = \{g \in \widehat{G} \mid g\eta(g) = 1\}/G.$$

For each connected component of this stack, containing $a \in \widehat{G}$, the isotropy group is the η-fixed point of the centralizer of a, denoted $\widehat{K}_a$. This is thought as the fixed point of the involution induced by the conjugation by $(a, \eta) \in \widehat{G} \rtimes \Gamma$. Let I be the et of connected components of $(B\widehat{G})^\Gamma$. Then,

$$(B\widehat{G})^\Gamma = \prod_{a \in I} B\widehat{K}_a,$$

and this actually coincides with the real forms appearing in Vogan's formulation of LLC.

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infinitesimal character

Conjecture 4.3 (Soergel's reformulation of Vogan's reformulation of archimedean Langlands)**.**

$$\bigoplus_{a\in I} D^b(\mathrm{Rep}_{G^a_{\mathbb{C}}}) = D^b((\widehat{B}\backslash\widehat{G}/\widehat{B})^\Gamma).$$

Similarly, this follows the pattern of Koszul duality, that the natural t-structure of the LHS is the Koszul dual t-structure of the RHS. ABV's approach is to patch $\widehat{K}_a\backslash\widehat{G}/\widehat{B}$ in a way that there is an interesting geometry (just like the geometry of Schubert cells), which is extremely delicate. Here we can avoid such delicacy. I am still not sure why ABV's parameter space is the right one to consider.

5. Loop spaces and D-modules

We now try to relate this more "folklore" conjecture to a more modern conjecture involving coherent sheaves over the moduli of Langlands parameters.

Slogan: perverse sheaves on X = D-modules on X = coherent sheaves on $\mathcal{L}X$.

Here, $\mathcal{L}X$ is the **free loop space** of X. Such notion makes sense in the world of derived stacks, as it combines stacks with topological spaces. For example, we let $S^1 = B\mathbb{Z}$, where $\mathcal{O}(S^1) = k[\epsilon]/\epsilon^2$. Then,

$$\mathcal{L}X = \mathrm{Map}(S^1, X).$$

A crucial observation is that

$$\mathcal{L}BG \cong G/G,$$

the adjoint quotient. This is because by definition of BG, $\mathcal{L}BG$ classifies G-local systems on S^1.

6. Koszul duality: perverse vs. coherent

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